



CHINSAPO CLUSTER MOCK

MALAWI SCHOOL CERTIFICATE OF EXAMINATIONS

AGRICULTURE

PAPER I (100 Marks)

(THEORY)

Date: ___ March, 2020

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 11 pages.
Please check.
2. This paper has **two** sections, **A** and **B**. Section **A** has 8 questions and section **B**, **three** questions.
3. Answer all questions in the spaces provided.
4. In the table provided on this page, tick against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
		TOTAL	

SECTION A (70 Marks)

1a (i) Define the term “inbreeding”

(1 mark)

(ii) State **one** advantage of “inbreeding”

(1 mark)

b. Give any **two** breeds of cattle for beef production.

(i) _____

(ii) _____

(2 marks)

2a. Give any **two** ways in which trading of agricultural commodities is important at international level.

(i) _____

(ii) _____

(2 marks)

b. Explain any **one** way that can facilitate trading of agricultural commodities from one district to another.

(2 marks)

c (i) State any **one** source of marketing costs.

(1 mark)

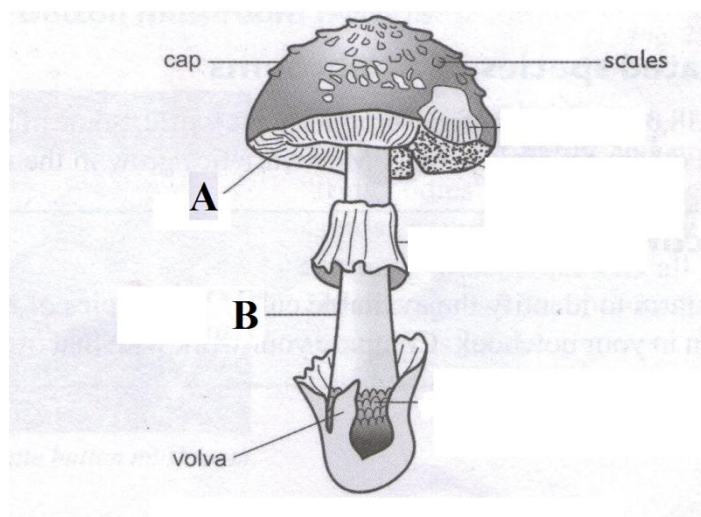
(ii) Explain the relationship between marketing costs and marketing margin.

(2 marks)

- (iii) Mr. Chimzimu sold maize to ADMARC which he bought from Malingunde farmers. ADMARC sold the maize to Likuni Boys Secondary School for consumption. Draw a marketing channel and name it.

(5 marks)

3. **Figure 1** is diagram of a mushroom. Use it to answer the questions that follow.



- a. Name the parts labelled **A**, **B** and **C** and state their functions.

(i) **A** _____

(ii) **Function** _____

(i) **B** _____

(ii) **Function** _____

(4 marks)

b. Mention any **two** factors to consider when selecting a site for mushroom production.

(i) _____

(ii) _____

(2 marks)

c. Explain any **one** way through which mushroom production can assist the development of the national economy.

(2 marks)

4a. State the **three** conditions that a nutrient must satisfy for it to be called an essential plant nutrient.

(i) _____

(1 mark)

(ii) _____

(1 mark)

(iii) _____

(1 mark)

b. State any **two** functions of Phosphorus.

(i) _____

(ii) _____

(2 marks)

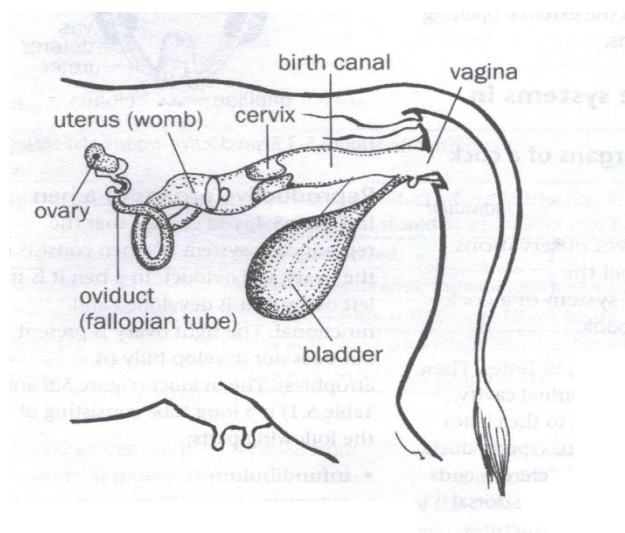
c. Describe any **one** nutrient deficiency symptom of phosphorus in crops.

(2 marks)

- d. A farmer has 2 hectares of land and require **35kg** of Phosphorus per hectare. He intends to use **23:21:0+4S** as a source of Phosphorus. Calculate the number of bags that will be required to apply.

(5 marks)

5. **Figure 2** is the reproduction system of a cow. Use it to answer the questions that follow.



- a. Which part is responsible for secreting sex hormones.

(1 mark)

- b. Mention any **one** function of the uterus.

(1 mark)

- c. Explain why it is recommended to give extra concentrates to a gestating cow three months before parturition.

(2 marks)

- 6a. State any **two** importance of irrigation.

(i) _____

(ii) _____

(2 marks)

- b. Explain the difference between surface and drip irrigation based on the following characteristics.

- (i) Water requirement

(2 marks)

- (ii) Economic requirement

(2 marks)

- (iii) Technical know - how

(2 marks)

- c. Mention any **one** problem that is associated with irrigation and state how it is solved.

(2 marks)

- 7a. Explain the following stages in a scientific approach to experimentation

- (i) Problem identification

(2 marks)

(ii) Data collection

(2 marks)

- b. Form 4 students intends to carry out an experiment to find the appropriate weeding frequency on a maize production. The class teacher advises them to use the following information.

A.

Treatments	Weeding frequency
1	Zero
2	Once
3	Two times
4	Three times
5	Four times

B. Random Number Table

4	8	1	6	5	9	3	7	2
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- (i) Using the random number table, design the experiment in three blocks.

(6 marks)

(ii) In the experiment state the control treatment.

(1 mark)

(iii) Give a reason for your answer in **b (i)**.

(1 mark)

8a. Give any **two** examples of natural pastures.

(1 mark)

b. State any **one** advantages of rotational grazing in pasture management.

_____ **(1 mark)**

c. Explain the hay making process in pasture conservation.

(6 marks)

SECTION B (30 Marks)

9. Explain any **five** advantages of agro based industries.

[illegible]

(10 marks)

FIRST NAME_____ SURNAME_____ CLASS____

10. With an aid of a well labelled diagram explain how you can carry out layering in plant propagation.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(10 marks)

11. Explain any **five** factors that affect the soil PH.

[illegible]

(10 marks)

END OF QUESTION PAPER.